

Introduction

Bond disorder in non-collinear antiferromagnets is a **non-trivial perturbation**. We describe -

- The effect of a single **bond defect** on a non-collinear magnetic order.
- Bond disorder** and the **stability** of the homogeneous ordering.

Single bond defect: emergent dipole

Triangular lattice (TLAF): Non-collinear Heisenberg magnet + 120° spiral order. A single **bond defect** relieves the frustration locally.

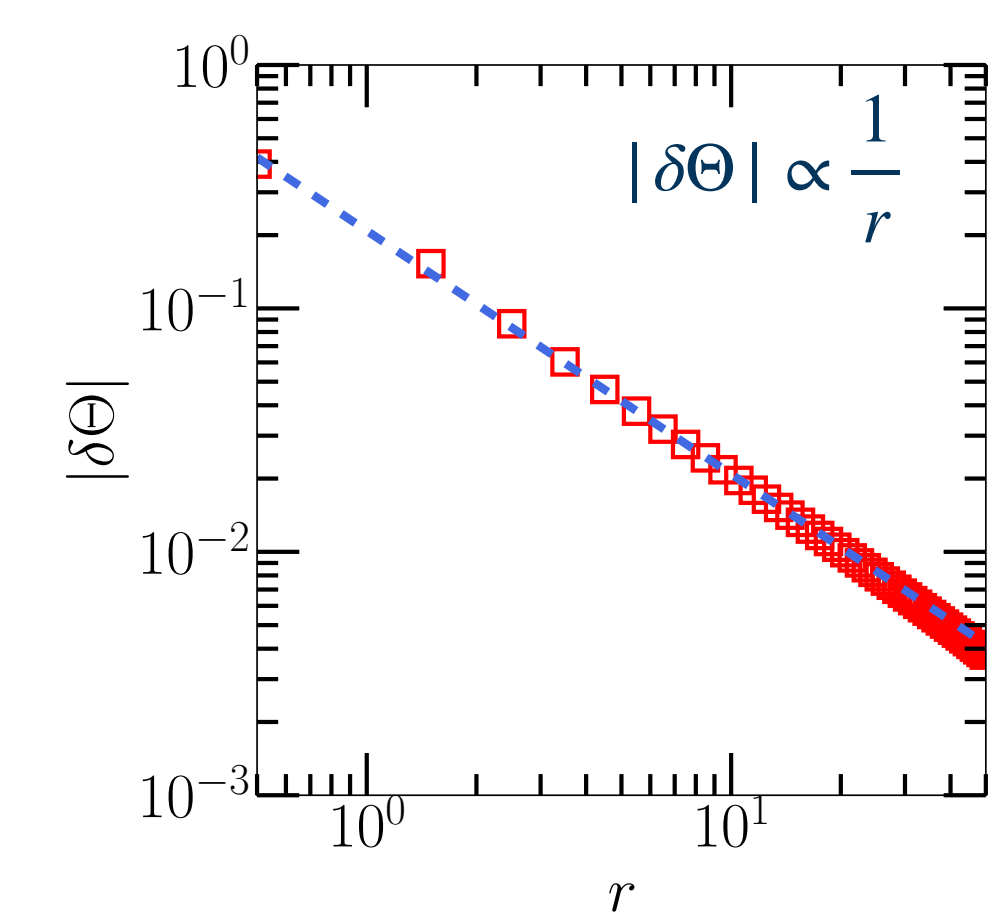
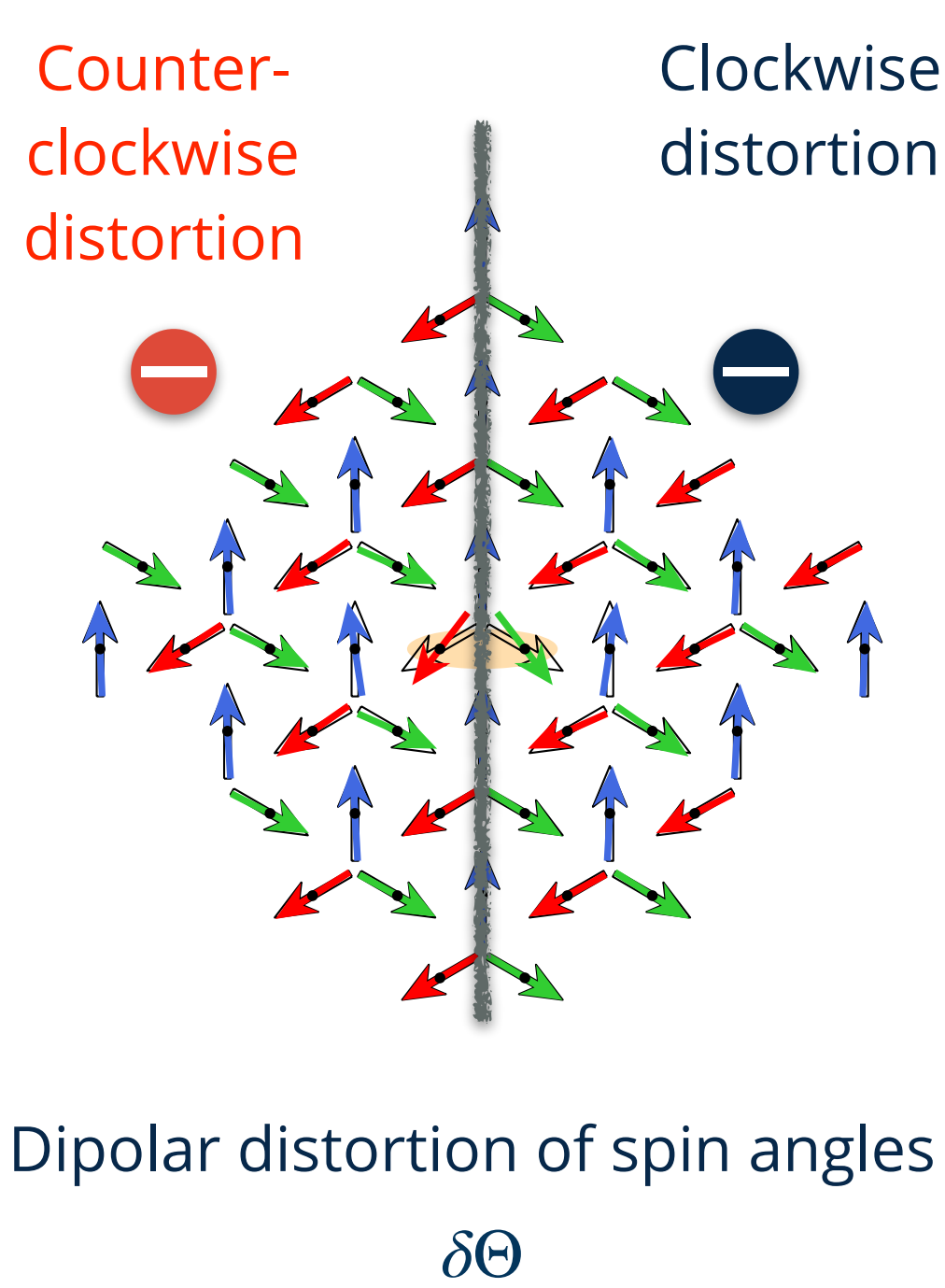
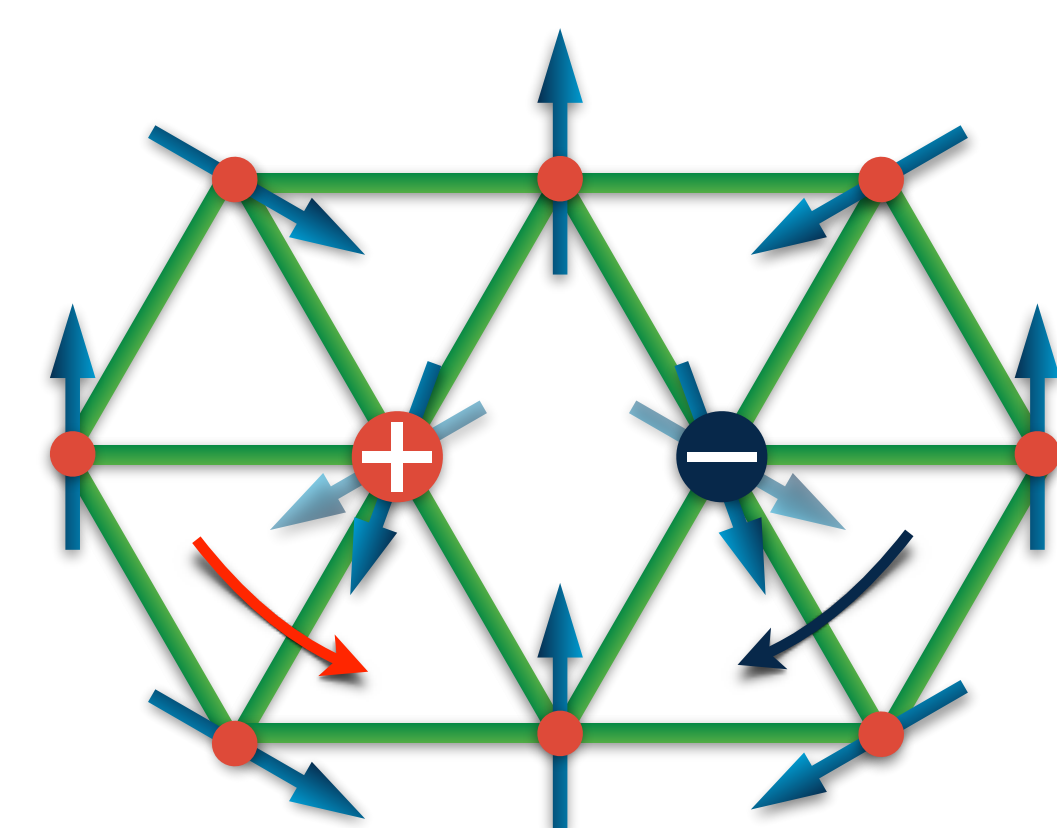
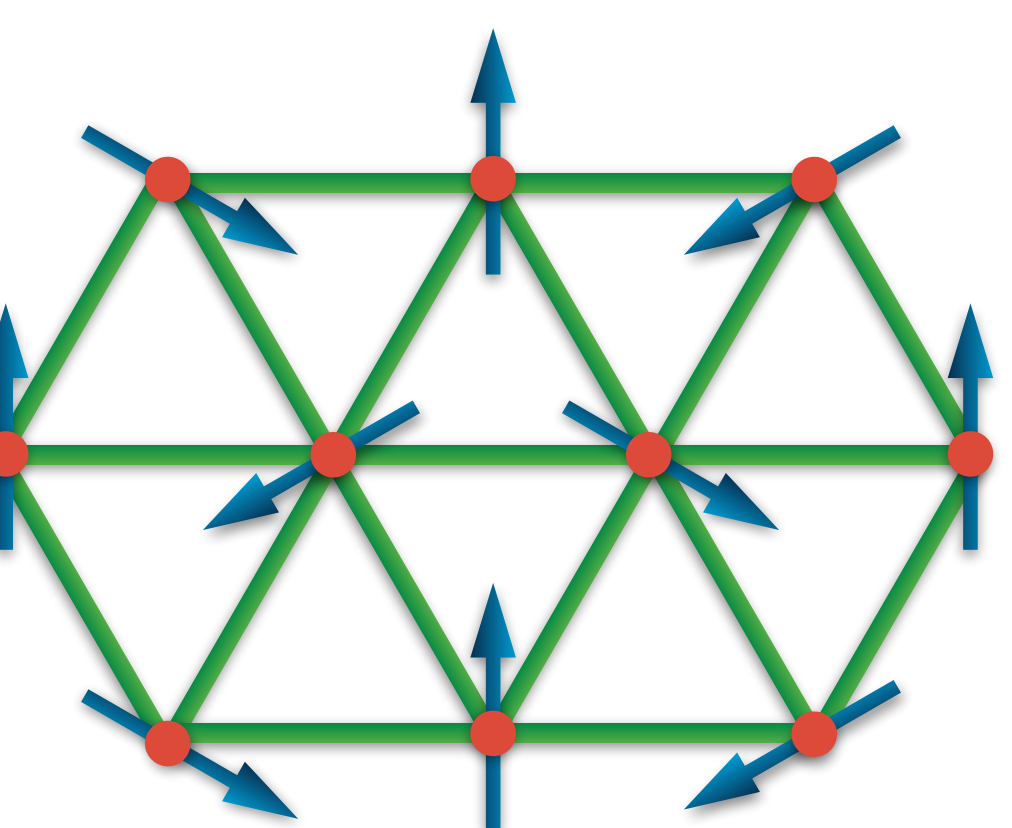
$$H = J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + \delta J \mathbf{S}_0 \cdot \mathbf{S}_1$$

Bond defect leads to a **dipolar response**.

- Spins reorient **anti symmetrically** with dipolar mirror symmetry.
- Magnitude of distortion falls off as $\frac{1}{r}$

Spins reorient anti symmetrically to become more collinear

Removal of a bond



Power law decay of angle distortion

Local dipolar field in body frame:

The defect induced perturbation:

$$\delta H_i \sim \delta J \left(S_i^x - S_{i+ae_{01}}^x \right)$$

Where ae_{01} is the bond vector.

In momentum space this leads to the **dipolar form factor** ($\mathbf{q} \cdot \mathbf{e}_{01}$).

$$\delta H_{\mathbf{q}} \sim \delta J (\mathbf{q} \cdot \mathbf{e}_{01}) S_{\mathbf{q}}^x$$

With in-plane goldstone modes -

$$\langle \langle S_{-\mathbf{q}}^x; S_{\mathbf{q}}^x \rangle \rangle \propto \frac{1}{q^2}$$

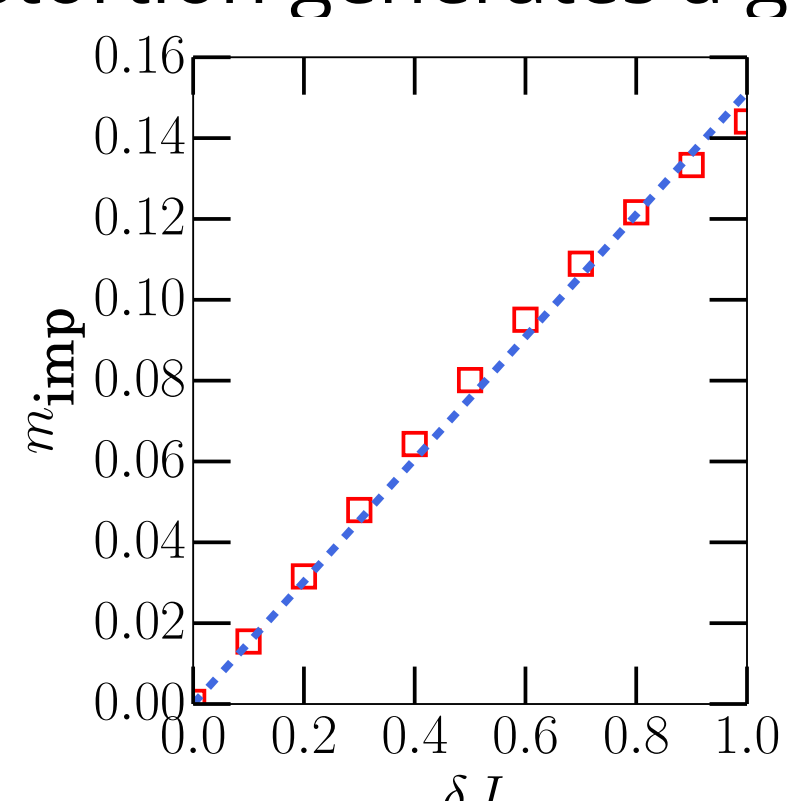
The angular distortion from the **linear-response** of the bond defect -

$$\delta \Theta(r) \propto \int d^d q e^{i\mathbf{q} \cdot \mathbf{r}} \frac{(\mathbf{q} \cdot \mathbf{e}_\alpha)}{q^2} \propto \frac{1}{r^{d-1}}$$

In $d = 2$ this relates the **p-wave symmetry** of bond-defect with $\frac{1}{r}$ **dipolar decay**.

Single bond defect: uniform moment

Spin-distortion generates a global uniform moment. In an $1/S$ expansion-



$$m_{tot} = \delta J S \left(0.39 - 0.08 S^{-1} + \mathcal{O}(S^{-2}) \right)$$

Bond-disordered non-collinear magnet: effective action

Spin-wave distortion $\delta \varphi$ on non-collinear ground state

For homogeneous bulk
Energy $\propto (\delta \varphi)^2$

Around the defective bonds
Energy $\propto \delta \varphi$

With sub-lattice α and bond vectors $\mathbf{e}_{\alpha\beta}$ - coarse-grained hamiltonian becomes -

$$H = J \sum_{x, \alpha \neq \beta} \mathbf{S}^\alpha(\mathbf{x}) \cdot \nabla^2 \mathbf{S}^\beta(\mathbf{x}) + \sum_{x, \alpha \neq \beta} \delta J^{\alpha\beta}(x) \mathbf{S}^\alpha(\mathbf{x}) \cdot \left(\mathbf{e}_{\alpha\beta} \cdot \nabla \right) \mathbf{S}^\beta(\mathbf{x})$$

Goldstone modes generated on top of ordered state,

$$\mathbf{S}^\alpha(\mathbf{x}) = R(\mathbf{x}) \cdot \mathbf{S}_0^\alpha$$

which leads to the **effective goldstone mode action** -

$$S = - \int d\tau \int d^d x \text{Tr} \left[P_\mu R^{-1} \nabla_\mu^2 R \right] - \int d\tau \int d^d x \delta J^{\alpha\beta}(x) e_{\alpha\beta}^i \text{Tr} \left[P_x T^3 (R^{-1} \nabla_i R) \right]$$

- For triangular lattice, $R \in SO(3) \simeq O(3)/Z_2$
- $P_\mu = \text{diag}(p_{1\mu}, p_{2\mu}, p_{3\mu})$, the stiffnesses of non-collinear ordering.
- $T^3 \in \text{Lie}[SO(3)]$ in-plane generator coupling the **goldstone modes** to the **random dipolar field**.

Infrared catastrophe

Collection of bond defects with disorder averaging,

$$\overline{\delta J^{\alpha\beta}(\mathbf{x}) \delta J^{\gamma\sigma}(\mathbf{x}')} = \Delta^2 \delta^{\alpha\beta; \gamma\sigma} \delta^{(d)}(\mathbf{x} - \mathbf{x}')$$

In $d = 2$, the spin-correlation function diverges -

$$\langle \overline{\mathbf{S}(\tau, \mathbf{x}) \cdot \mathbf{S}(\tau', \mathbf{x}') } \rangle \propto \Delta^2 \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \rightarrow \infty$$

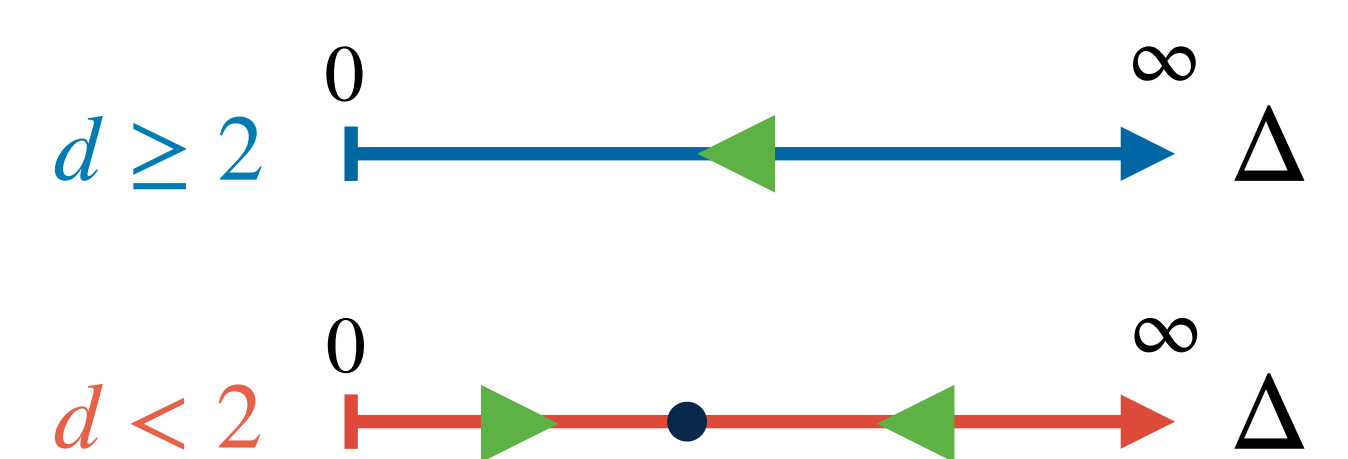
- For **non-vanishing disorder** strength Δ this would indicate the **breakdown of long range order** (LRO) [2]!
- However, $[\Delta] = (2 - d)/2$ is **marginal** in tree level in $d = 2$ and perturbative corrections are necessary.

Perturbative renormalization

With gaussian disorder distribution a **joint expansion** in $d - 2$ and Δ with the **replica method** -

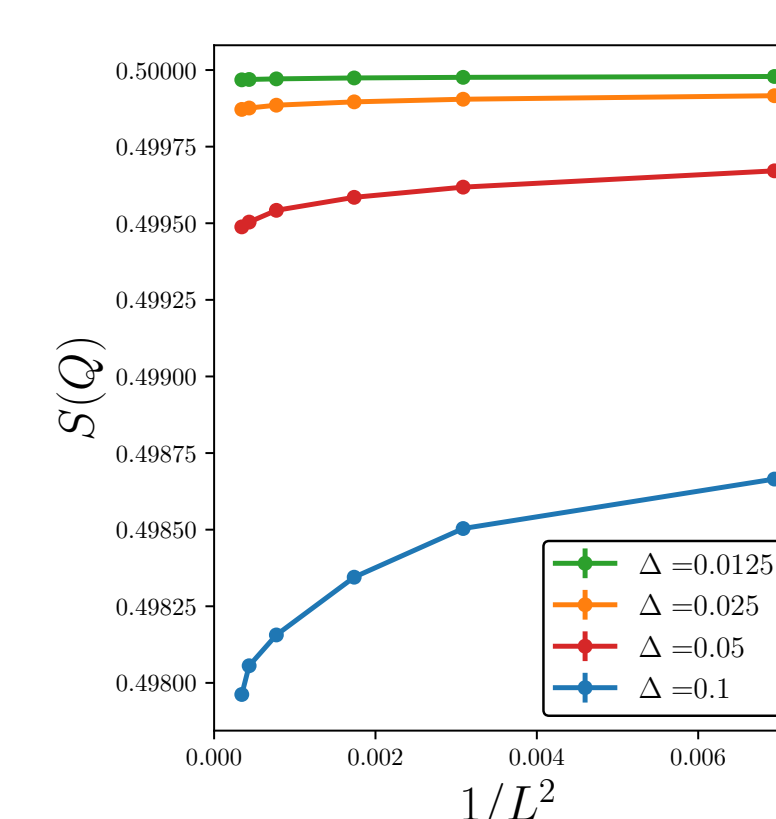
Up to two-loop order :

$$\beta(\Delta^2) = (d - 2)\Delta^2 + \frac{\Delta^4}{24\pi}$$

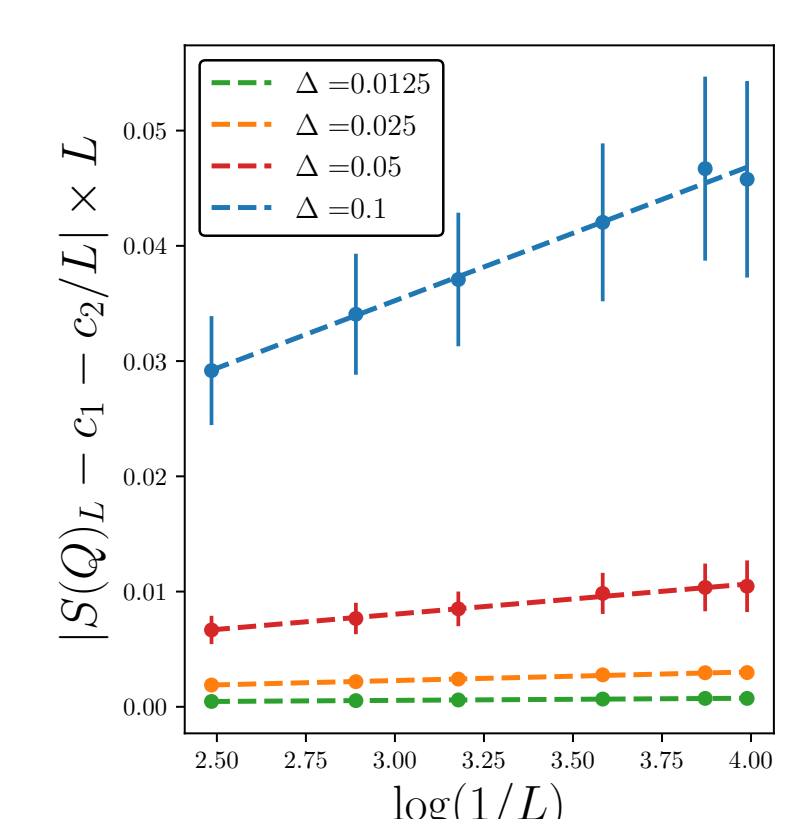


- Disorder is a **marginally irrelevant** perturbation in $d \geq 2$.
- So two-dimensional non-collinear LRO is **robust** against **weak disorder** [3].

Marginal disorder: numerical evidence (TLAF)



Survival of LRO for weak disorder



Sub-leading scaling for marginal behavior

References

- [1] P. Azaria et. al, Phys. Rev. Lett. **68** (1992) and Nucl. Phys. **B408**, 485 (1993), T. Dombre and N. Read, Phys. Rev. B **39**, 6797 (1989)
- [2] Y. Imry and S. Ma, Phys. Rev. Lett. **35**, 1399 (1975), A. Aharony, Solid State Comm. **28**, 667 (1978)
- [3] Wu. et. al, arXiv:1802.00935, Watanabe et. al, J. Phys. Soc. of Japan **83**, 034714 (2014)